

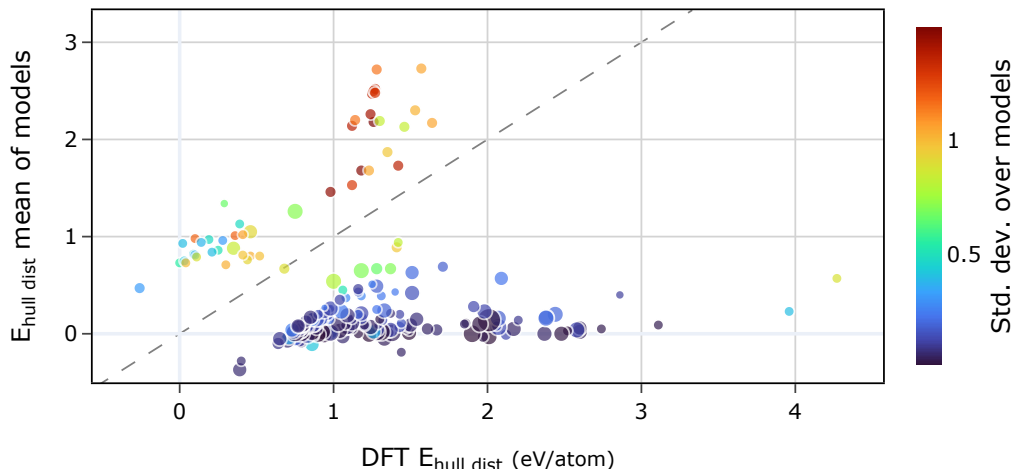
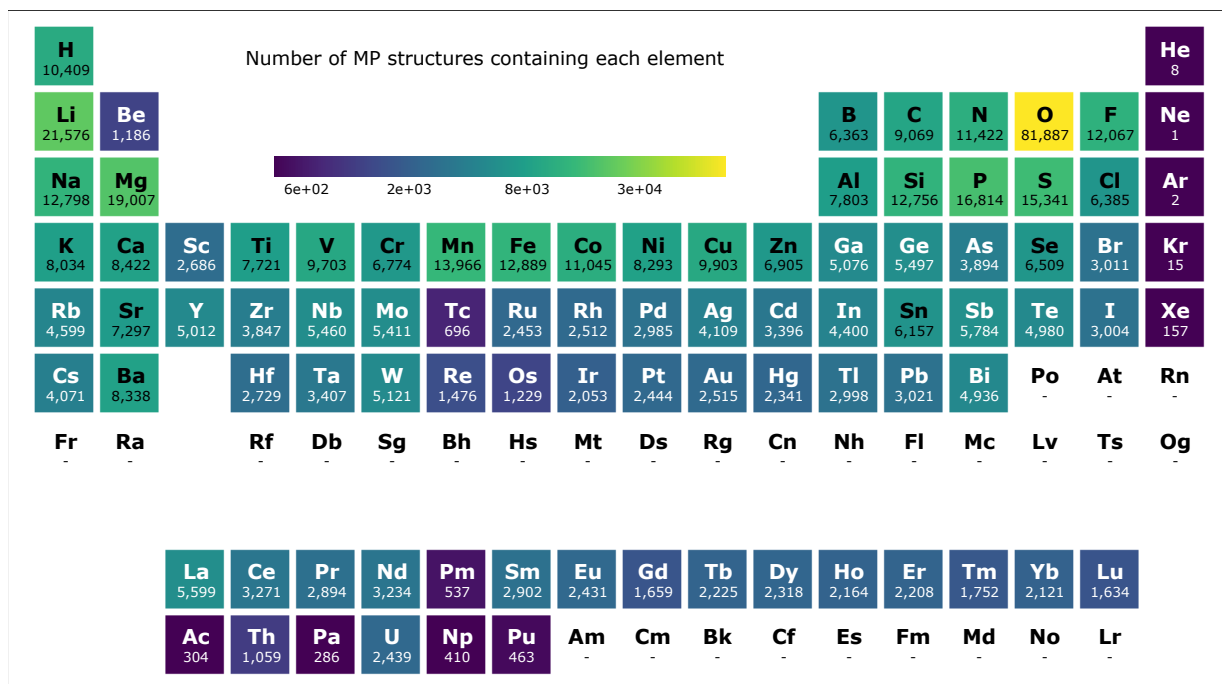


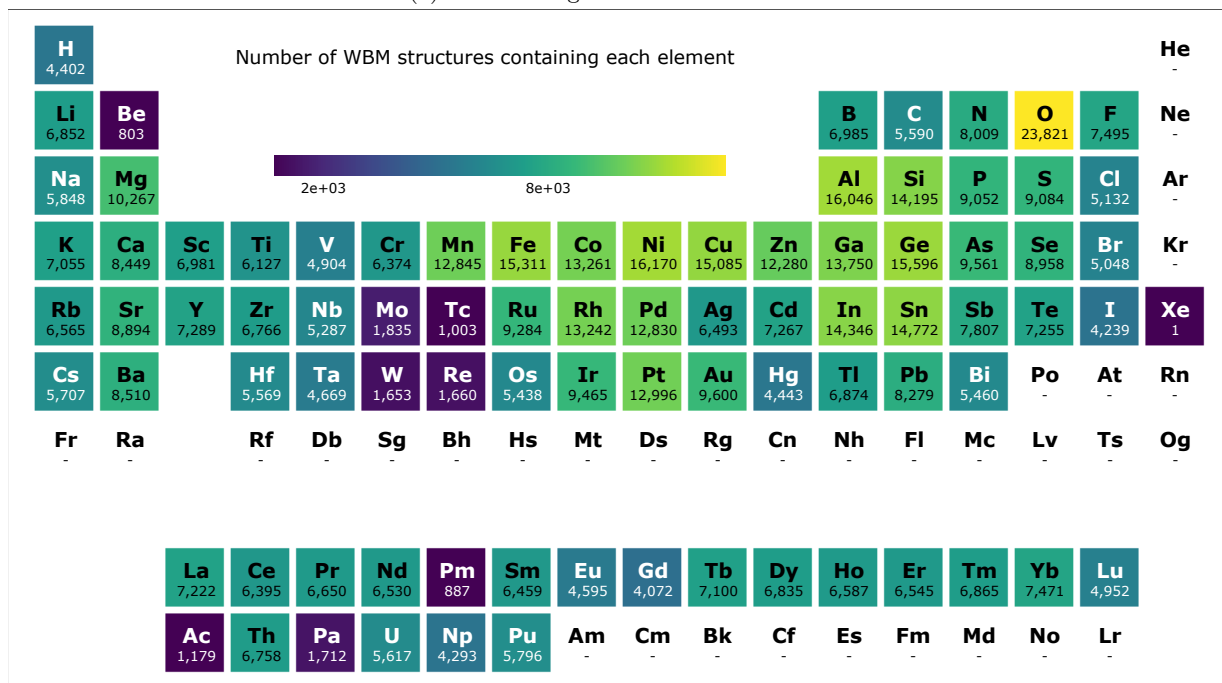
Figure S7: DFT vs predicted hull distance (average over all models) for the 200 largest error structures. Points are colored by model disagreement as measured by the standard deviation in hull distance predictions from different models. Point scale with the number of atoms in the structures. This plot shows that high-error predictions are biased towards predicting too small hull distances. This is unsurprising considering MP training data mainly consists of low-energy structures. There is a strong color separation between the mostly dark blue low-energy bias predictions and the high-error green or red predictions. Blue means models are in good agreement, i.e. all models are wrong together. Red means large-error predictions with little model agreement, i.e. all models are wrong in different ways. Some of the blue points with large errors yet good agreement among models may be accurate ML predictions for a DFT relaxation gone wrong. The dark blue points also tend to be larger corresponding to larger structures where DFT failures are less surprising. This suggests ML model committees might be used to cheaply screen large databases for DFT errors in a high-throughput manner.

most 10 eV/atom vs less than 1 eV/atom spread in the hull distances. This highlights why stability prediction is a much more challenging task than predicting energy of formation. It requires correctly ranking the subtle energy differences between chemically similar compounds in the same chemical system rather than comparing a single material with the reference energies of its constituent elements. DFT has been shown to significantly benefit from the systematic cancellation of errors between chemically similar systems when trying to identify the lowest-lying polymorph [83]. This beneficial cancellation has yet to be conclusively demonstrated for ML stability predictions. So far, only the lack thereof has been shown in [14] where they encountered a much more random error distribution among similar chemistries than simulations from first principles.

Finally, Believing MPtrj to be an influential dataset for the near-term continued development of universal inter-atomic potentials, we plot histograms showing the distributions of target values for energies, forces, stresses and magnetic moments in fig. S12. The histogram in fig. S12e shows the distribution of the number of sites in MPtrj structures. The inset displays the same histogram log-scaled y-axis as well as a cumulative line to show that 90% of MPtrj structures contain fewer than 70 sites. fig. S13 shows the forces and magnetic moments for the MPTrj dataset projected onto different elements.

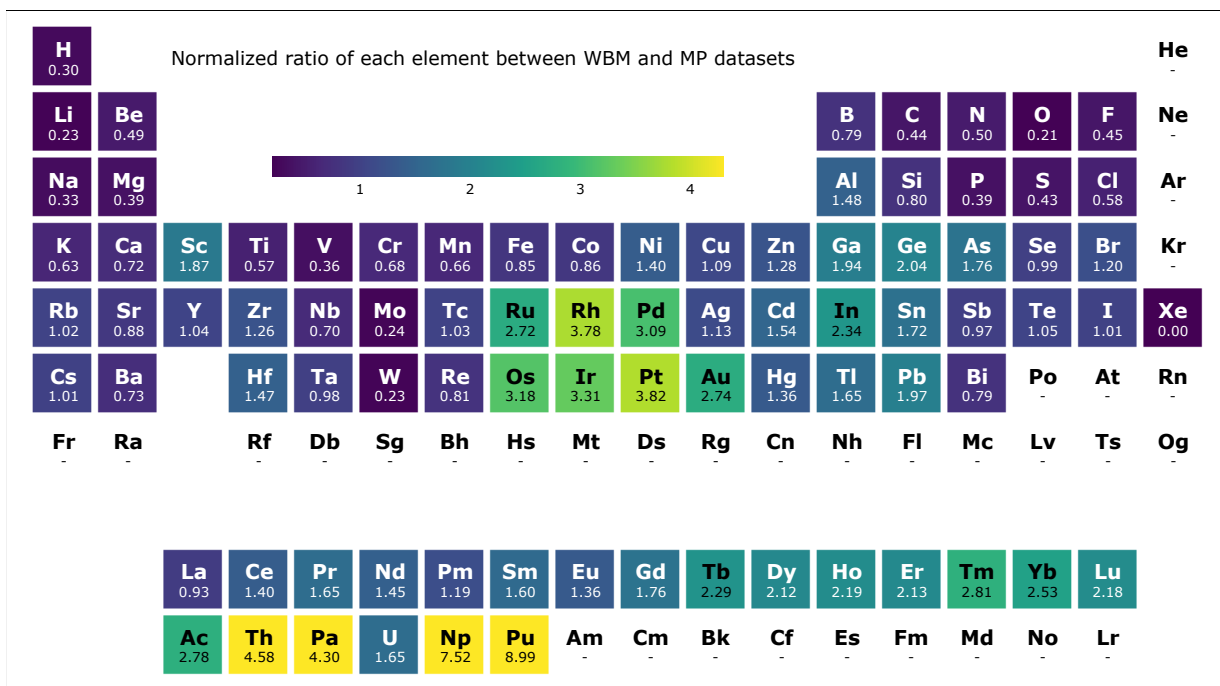


(a) MP training set element occurrence

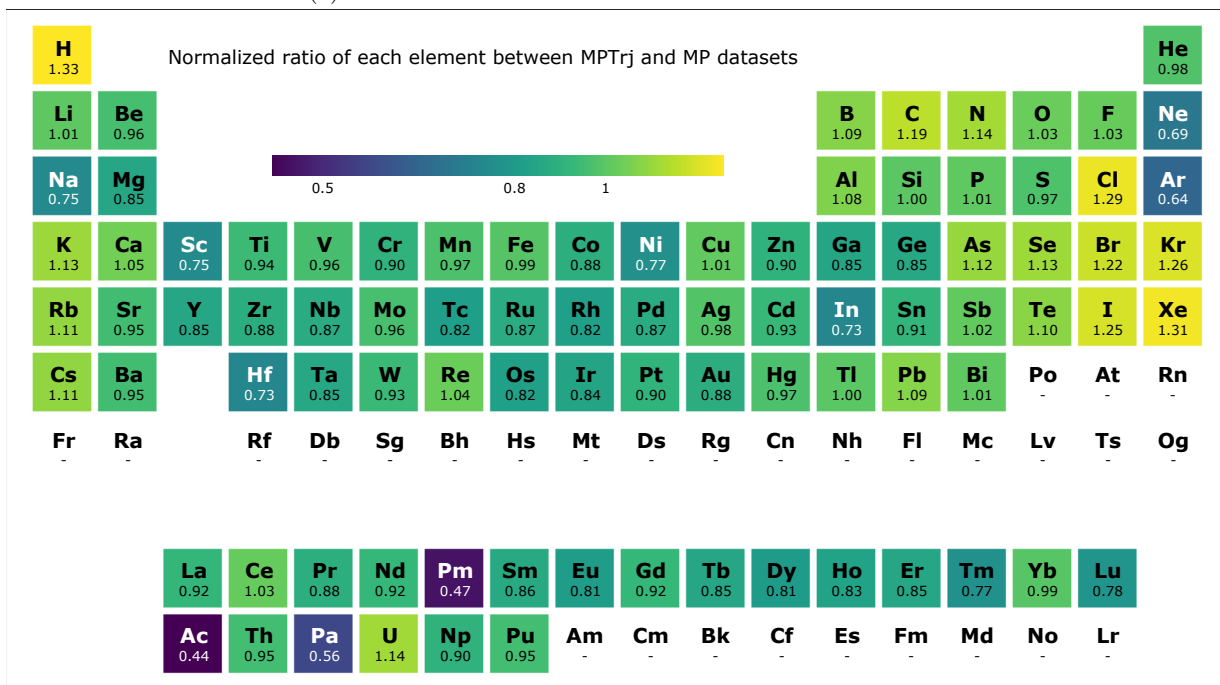


(b) WBM test set element occurrence

Figure S8: The number of structures containing a given element in the MP training set and WBM test set [40]. The WBM test set in relative terms contains noticeably fewer oxides than MP (and, by extension, MPtrj) with just 11% rather than 53% of structures containing oxygen. Made with pymatviz [84].



(a) Normalized ratio of elements in WBM dataset to MP.



(b) Normalized ratio of elements in MPTrj dataset to MP.

Figure S9: **a)** shows the ratio of elements in the WBM test set to the MP training set. The figure makes clear that WBM explores a distinct chemical space to MP with noble-transition metals, post-transition metals, lanthanides, actinides and metalloids seen more frequently in WBM than MP. Similarly, **b)** shows the ratio of elements in the MPTrj dataset to the MP training set. We note a slight overabundance of structures containing hydrogen and halides, indicating that more frames were selected from structures containing these elements which might correlate with the number of ionic steps to find their ground states.

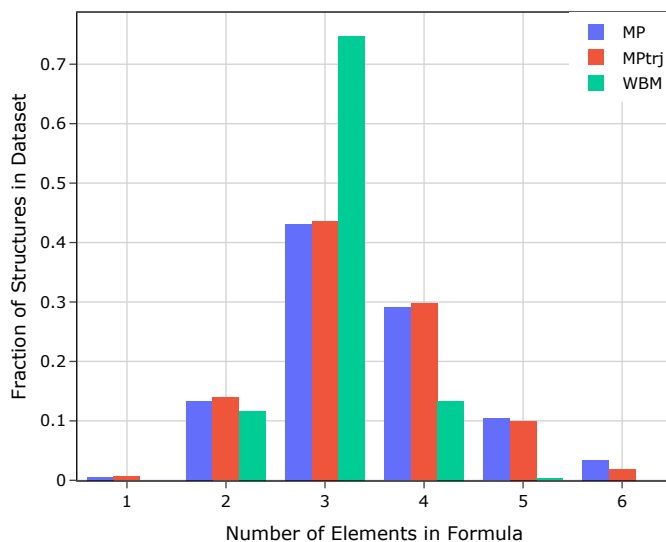


Figure S10: Distributions of unique elements per structure in MP, MPtrj and the WBM test set. The bar heights are normalized by the total number of structures in each dataset. WBM is dominated by ternary phases making up 75% of the dataset followed by about 13% for quaternaries and 12% for binaries. MP has a more even distribution, in particular with more than double the relative share of quaternary phases and a significant number of quaternaries which are almost absent from WBM. Not shown in this plot for visual clarity are 3% of MP structures containing more than 5 elements (up to 9). We also include MPtrj in this plot to show a slight drop in the relative abundance of quaternaries and higher phases vs MP ground states. This may be due to a poor choice of convergence criteria in early MP relaxation workflows that scaled with the size of the structure (see `EDIFF_PER_ATOM` parameter in `pymatgen` VASP input sets), resulting in unconverged large structures with short relaxation trajectories entering the database. Short relaxations would result in fewer frames of such structures selected for MPtrj. This assumes structures of higher arity correlate with larger structures.

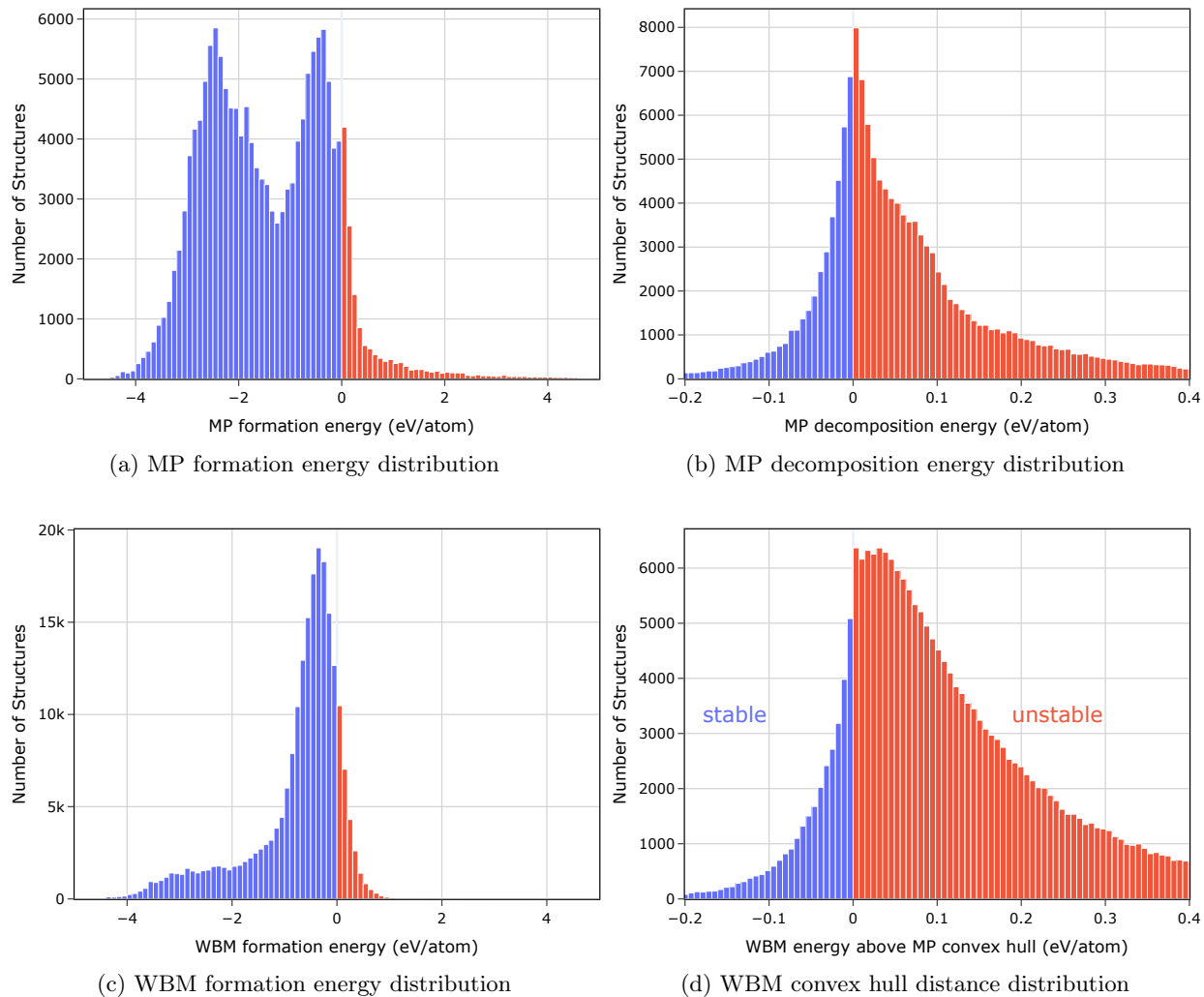


Figure S11: Distribution of formation energies, decomposition energy and convex hull distances for the MP training set and the WBM test set. In (a) the bimodality in the MP formation energy distribution is due to the MP anion correction scheme [85] which significantly lowers some formation energies, especially for oxides. The decomposition energy shown in (b) calculated as defined in [14]. We note that multiple materials with the same composition can have negative decomposition energies hence the number with negative decomposition energies is $\sim 43\text{k}$ compared to $\sim 35\text{k}$ that are on the hull. Looking at WBM in (c) and (d), we see that the distribution of convex hull distances is much tigher due to the change of reference and therefore a more discriminative benchmarking task in terms of goodness-of-fit measures like the coefficient of determination.